

Partial Manuscript - A survey on Positional Encodings

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1 Absolute Positional Embeddings

An initial method [Vaswani et al., 2017] to enhance vector representations of the tokens with positional information was to directly add positional vectors to the input. Thus for every input token in the sequence $u_1, u_2, \dots, u_T \in \mathbb{R}^d$ we add T positional encodings $p_1, p_2, \dots, p_T \in \mathbb{R}^d$ ($u_i^{PE} = u_i + p_i$). The authors proposed a time-correlated relationship between them through sinusoids such that the model can attend relative positions.

Thus for a token u at position k and dimension $i \in \{1, 2, \dots, d\}$:

$$p_{k,i} = \begin{cases} \sin\left(\frac{k}{10000^{i/d}}\right) & \text{if } 2 \mid i \\ \cos\left(\frac{k}{10000^{i/d}}\right) & \text{otherwise} \end{cases}, p_{k,i} \in \mathbb{R} \quad (1)$$

2 Relative Position Embeddings

Extending the original Transformer's, [Shaw et al., 2018] replaces absolute positional representations considering local distances between the tokens as a signal of information. Recalling the self-attention between two tokens u_i and u_j (at positions i and j respectively in the sequence $i, j \in \{1, 2, \dots, T\}$) as the normalized inner product of Q_i and K_j :

$$Att(u_i, u_j) = softmax\left(\frac{Q_i^T K_j}{\sqrt{d}}\right) V_j \quad (2)$$

where q and k are linear projections of the token representations

$$Q_i = W^Q u_i \quad K_j = W^K u_j \quad V_j = W^V u_j \quad (3)$$

$$W^Q, W^K, W^V \in \mathbb{R}^{d \times d} \text{ are time-constant trainable parameters} \quad (4)$$

we modify it by introducing two trainable biases $\beta^K, \beta^V \in \mathbb{R}^{2T-1 \times d_h}$ ($\max T$ is fixed, d_h is the head dimension, here $d = d_h$), so the attention is:

$$Att(u_i, u_j) = softmax\left(\frac{(W^Q u_i)^T (W^K u_j + \beta_{j-i+T}^K)}{\sqrt{d}}\right) (W^V u_j + \beta_{j-i+T}^V) \quad (5)$$

Clipping the relative distance

The authors claim that precise relative position across large distances is useless, so they clip the relative distance $j - i$ to a clipping value k resulting in a smaller set of biases $\beta^K, \beta^V \in \mathbb{R}^{2k+1 \times d_h}$ and for a pair u_j and u_i the used bias is $\beta_{clip(j-i, -k, k)+k+1}^*$.

3 Rotary Positional Embeddings

In Rotary Positional Embeddings (RoPE), proposed in [Su et al., 2024], we assume $2|d$ such that we can split q_i in a number of $d/2$ 2-dimensional parts $Q_i = [q_{i,[1,2]}; q_{i,[3,4]}; \dots; q_{i,[d/2-1,d/2]}]$ and k_j respectively. RoPE rotates each individual 2D vector using a rotation matrix R^{q_i} , parameterized by a wavelength θ (authors use a default value of 10000), the index of q in the query vector Q and the position of the token in the sequence.

$$R^{q_i,[c,c+1]} = \begin{bmatrix} \cos i\Theta_{c/2} & -\sin i\Theta_{c/2} \\ \sin i\Theta_{c/2} & \cos i\Theta_{c/2} \end{bmatrix}, \text{ where } \Theta_\gamma = \theta^{-2(\gamma-1)/d} \quad (6)$$

So a rotated query Q (or key K) can be obtained by multiplying the rotation matrix $R^{Q_i} = \text{diag}(R^{q_i,[1,2]}, R^{q_i,[3,4]}, \dots, R^{q_i,[d/2-1,d/2]}) \in \mathbb{R}^{2 \times 2} \in \mathbb{R}^{d \times d}$ with the linear projection of the token u_i :

$$Q_i^{RoPE} = \begin{bmatrix} \cos i & -\sin i & 0 & 0 & \dots & 0 & 0 \\ \sin i & \cos i & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & \cos i\theta^{-2/d} & -\sin i\theta^{-2/d} & \dots & 0 & 0 \\ 0 & 0 & \sin i\theta^{-2/d} & \cos i\theta^{-2/d} & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & \cos i\theta^{-1} & -\sin i\theta^{-1} \\ 0 & 0 & 0 & 0 & \dots & \sin i\theta^{-1} & \cos i\theta^{-1} \end{bmatrix} W^Q u_i \quad (7)$$

The entire formula for computing the attention weights can be reformulated such that we use a single rotation matrix:

$$\text{Att}(u_i, u_j) = \text{softmax}\left(\frac{(R^{Q_i} W^Q u_i)^T (R^{K_j} W^K u_j)}{\sqrt{d}}\right) = \text{softmax}\left(\frac{W^Q u_i R^{*i-j} W^K u_j}{\sqrt{d}}\right) \quad (8)$$

This rotation can be reformulated in a more efficient manner by mapping the 2-dimensional parts $q_{i,[1,2]}, q_{i,[3,4]}, \dots, q_{i,[d/2-1,d/2]} \in \mathbb{R}^2 \rightarrow \mathbb{C}^2$ (odd-indexed dimensions are the real parts, and real-indexed dimensions are the imaginary parts for each pair, below denoted using the ket notation for the entire vector of dimension d). In the complex plane, the rotation matrix $R^{*i,[c,c+1]} \in \mathbb{R}^{2 \times 2}$ is replaced by a complex number $e^{i\theta} = \cos\theta + i\sin\theta$ (a counter-clockwise rotation of the vector by θ radians), so:

$$Q_m^{\mathbb{C}RoPE} = \begin{bmatrix} \Theta_1 & 0 & \dots & 0 & 0 \\ 0 & \Theta_2 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & \Theta_{d/2-1} & 0 \\ 0 & 0 & \dots & 0 & \Theta_{d/2} \end{bmatrix} |W^Q u_m\rangle_{\mathbb{C}}, \text{ where } \Theta_\gamma = e^{i\theta^{-2(\gamma-1)/d}} \quad (9)$$

$$Q_{m,k}^{RoPE} \leftarrow |Q_{m,k}^{RoPE}\rangle_{\mathbb{R}} = \begin{cases} \text{Re}(Q_{m,k}^{\mathbb{C}RoPE}) & \text{if } 2 \nmid k \\ \text{Im}(Q_{m,k}^{\mathbb{C}RoPE}) & \text{if } 2 \mid k \end{cases} \text{ for } k = 1, 2, \dots, d \quad (10)$$

Similarly, we can reformulate such that we use a single rotation matrix in the complex plane:

$$\text{Att}(u_n, u_m) = \text{softmax}\left(\frac{|(e^{in\theta} |W^Q u_n\rangle_{\mathbb{C}})^T (e^{im\theta} |W^K u_m\rangle_{\mathbb{C}})\rangle_{\mathbb{R}}}{\sqrt{d}}\right) \quad (11)$$

$$= \text{softmax}\left(\frac{||W^Q u_i\rangle_{\mathbb{C}} e^{i(n-m)\theta} |W^K u_j\rangle_{\mathbb{C}}\rangle_{\mathbb{R}}}{\sqrt{d}}\right) \quad (12)$$

Extending the context window

A simple way (with inherent drawbacks) to allow larger contexts is to scale the sequence position parameter m of the rotation matrix $R^{*,m}$ by the newer max sequence length $m \leftarrow mL_{old}/L_{new}$ ($L_{new} > L_{old}$) along with finetuning until the model gets used to the newer compressed positional embeddings, while holding the same base frequency.

Instead of scaling the frequencies equally, the NTK-Aware interpolation considers scaling the frequencies differently:

$$\Theta_\gamma = (\theta s^{\frac{d}{d-2}})^{-\gamma/d} \text{ or } e^{i(\theta s^{\frac{d}{d-2}})^{-\gamma/d}} \text{ for rotations in } \mathbb{C} \quad (13)$$

There are also extended versions of such strategies like NTK-by-parts interpolation and YaRN Peng et al. [2023], but recalling their functionality is outside the scope of this survey.

4 Fixing and improving RoPE

As seen from the experiments, while low frequencies in standard RoPE are used for semantic information, they are not robust over long contexts because rotations, even at low frequencies, can eventually lead to misalignment.

p -RoPE was proposed to provide robust semantic channels that are distance-agnostic with increasing base wavelength. The intuition is that if the lowest frequencies are indeed problematic for robustness over large contexts, truncating them should not harm them and might even improve performance by allowing RoPE to have semantic channels that do not change with relative distance. The modification aims to address this by creating truly distance-agnostic semantic channels. It achieves this by applying RoPE rotations to only a fraction p of the $d/2$ available 2-dimensional parts, typically those corresponding to higher frequencies, while the remaining $(1 - p)$ fraction (lowest frequencies) are not rotated (in their experiments, $p = 0.75$ was found to be the best). The number $n \leq d/2$ of 2D parts that will undergo RoPE rotation is determined by p :

$$n = \lfloor p \cdot |Q_i| \rfloor, \text{ where } Q_i = [q_{i,[1,2]}; q_{i,[3,4]}; \dots; q_{i,[d/2-1,d/2]}] \quad (14)$$

The remaining parts $d/2 - n$ will not be subject to rotation. These parts are chosen to be those associated with the lowest frequencies. Let S_{RoPE} be the set of indices for 2D parts that are rotated, and S_{NoPE} be the set of indices for parts that are not rotated. If we assume frequencies Θ_γ are ordered such that $\Theta_1 > \Theta_2 > \dots > \Theta_{d/2}$ (i.e., $\gamma = 1$ is the highest frequency, $\gamma = d/2$ is the lowest), then:

$$S_{RoPE} = \{\gamma \mid 1 \leq \gamma \leq n\} \quad S_{NoPE} = \{\gamma \mid n < \gamma \leq d/2\} \quad (15)$$

For $m \in S_{NoPE}$, the term is simply a standard dot product for that 2D part, providing no rotational positional information, thus acting as a NoPE Haviv et al. [2022] channel for those dimensions.

- If $p = 0$, then $n = 0$, $S_{RoPE} = \emptyset$, and $S_{NoPE} = \{1, \dots, d/2\} \Rightarrow \text{NoPE}$
- If $p = 1$, then $n = d/2$, $S_{RoPE} = \{1, \dots, d/2\}$ and $S_{NoPE} = \emptyset \Rightarrow \text{RoPE}$

References

- Adi Haviv, Ori Ram, Ofir Press, Peter Izsak, and Omer Levy. Transformer language models without positional encodings still learn positional information. *arXiv preprint arXiv:2203.16634*, 2022.
- Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. Yarn: Efficient context window extension of large language models. *arXiv preprint arXiv:2309.00071*, 2023.
- Peter Shaw, Jakob Uszkoreit, and Ashish Vaswani. Self-attention with relative position representations. *arXiv preprint arXiv:1803.02155*, 2018.
- Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.